

Regression Model Evaluation Report

Project: Healthcare Cost Estimator

Models Trained

The recorded comparison includes Linear Regression, Random Forest, XGBoost, and Tuned XGBoost.

Evaluation Metrics

Model	MAE	RMSE	R2
Linear Regression	4176.9200	5956.4500	0.8069
Random Forest	2619.6100	4665.0500	0.8816
XGBoost	2592.2500	4456.0000	0.8919
Tuned XGBoost	2444.1600	4229.8000	0.9026

Loaded Model Verification on Current Test Split

MAE	MSE	RMSE	R2
2444.1596	17891209.7306	4229.8002	0.9026

Selected Best Model

Best model by highest recorded R2 score: Tuned XGBoost with R2=0.9026, MAE=2444.16, and RMSE=4229.80.